

Revisiting Discriminative vs. Generative Classifiers: Theory and Implications

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Introduction: Motivation

Problem Statement

- ▶ Large pre-trained deep models are widely used for transfer learning
- ▶ Linear evaluation freezes model parameters and trains a simple linear classifier
- ▶ Default choice is logistic regression, but **its statistical efficiency may be suboptimal**, especially with few labeled samples

Introduction: Goal of the Study

What the Authors Aim to Do

- ▶ Analyze **multiclass Naive Bayes vs logistic regression** in linear evaluation
- ▶ Understand **sample efficiency**: how many labeled examples are needed to reach optimal performance

Why This Matters

- ▶ Choosing a more efficient classifier can significantly improve performance in low-data regimes

Preliminaries: Notation and Setup

Notations

- ▶ X — domain set, $Y = \{1, \dots, K\}$ — label set, feature dimension n .
- ▶ H — set of functions $h : X \times Y \rightarrow \mathbb{R}^K$, prediction: $\arg \max_y h_y(x)$.
- ▶ Linear hypothesis family:

$$H_{\text{lin}} = \{h : h_y(x) = \langle w_y, x \rangle + b_y, \|w_y\|_2 \leq W, |b_y| \leq B, \forall y\}.$$

- ▶ Generalization error and minimal generalization error for loss ℓ :

$$R_\ell(h) = \mathbb{E}_{(x,y) \sim D}[\ell(h(x), y)], \quad R^* = \inf_{h \in H} R_\ell(h).$$

Discriminative vs. Generative Classifiers

Logistic Regression (Discriminative)

- ▶ Parameters: $\{w_y\}_{y=1}^K, \{b_y\}_{y=1}^K$.
- ▶ Prediction: $\hat{y} = \arg \max_y (\langle w_y, x \rangle + b_y)$.

Naive Bayes (Generative)

- ▶ Discrete inputs (with Laplace smoothing):

$$\hat{p}(x_i = 1 \mid y = k) = \frac{\#\{x_i = 1, y = k\} + \alpha}{\#\{y = k\} + 2\alpha},$$

$$\hat{p}(y = k) = \frac{\#\{y = k\} + \alpha}{m + K\alpha}.$$

- ▶ Continuous inputs: $p(x_i \mid y = k) \approx \mathcal{N}(\mu_{ki}, \sigma_i^2)$, with σ_i^2 independent of k (to keep decision boundary linear).
- ▶ Prediction: $\hat{y} = \arg \max_y \hat{p}(y) \prod_i \hat{p}(x_i \mid y)$.

Multiclass setup and Naive Bayes objects

Setup

- ▶ $h_{\text{Gen},m}, h_{\text{Dis},m}$ — hypotheses learned from m i.i.d. samples by naive Bayes and multiclass logistic regression.
- ▶ $h_{\text{Gen},\infty}, h_{\text{Dis},\infty}$ — their asymptotic limits.
- ▶ Goal: bound the finite-sample gaps

$$R_{0-1}(h_{\text{Gen},m}) - R_{0-1}(h_{\text{Gen},\infty}), \quad R_{0-1}(h_{\text{Dis},m}) - R_{0-1}(h_{\text{Dis},\infty}).$$

Definition

For classes k_1, k_2 define

$$\Delta a_{\text{Gen}}(x, k_1, k_2) = a_{\text{Gen}}(x, k_1) - a_{\text{Gen}}(x, k_2)$$

where $a_{\text{Gen}}(x, k) = \sum_{i=1}^n \log \hat{p}(x_i \mid y = k) + \log \hat{p}(y = k)$.

Main multiclass bounds, assumptions and sample complexity

Definition

Define

$$\tilde{G}(\tau) = \max_{k_1, k_2} \mathbb{P}_{(x, y) \sim \mathcal{D}} (|\Delta a_{Gen, \infty}(x, k_1, k_2)| \leq \tau n).$$

This is the probability that two class scores are nearly tied.

Theorem (Generalization bound)

Under mild assumptions, with probability at least $\geq 1 - \delta$:

$$R_{0-1}(h_{Gen, m}) \leq R_{0-1}(h_{Gen, \infty}) + \frac{K(K-1)}{2} \left(\tilde{G}\left(O\left(\sqrt{\frac{1}{m} \log\left(\frac{n}{\delta}\right)}\right)\right) + \delta \right).$$

If $\tilde{G}(\tau)$ is small for small τ , the finite-sample gap is small.

Bound on $\tilde{G}$ and result for Naïve Bayes

Assumptions

$$\forall k_1, k_2 (k_1 \neq k_2), \forall k \in \mathcal{Y} : \left| \sum_{i=1}^n (D_{kl}(p(x_i|y=k) \| p(x_i|y=k_1)) - D_{kl}(p(x_i|y=k) \| p(x_i|y=k_2))) \right| = \Omega(n),$$

The total signal increases proportionally to n .

$$\forall k_1, k_2 (k_1 \neq k_2), \forall k \in \mathcal{Y} : \mathbb{D} \left[\sum_{i=1}^n \log \frac{p(x_i|y=k_1)}{p(x_i|y=k_2)} \middle| y = k \right] = O(n^r), \quad r \in [1, 2).$$

The total variation does not grow too quickly.

Theorem (Sample complexity of Naïve Bayes)

If the assumptions hold, and for $\varepsilon_0 \in (0, 1)$, $\delta_0 \in (0, \frac{\varepsilon_0}{K^2}]$, $\tau < \zeta$, $\alpha = O(n^{r-1})$, $\zeta = \Omega(1)$, $\tilde{G}(\tau) \leq \frac{\alpha}{(\tau - \zeta)^2 n}$. To achieve

$$R_{0-1}(h_{Gen,m}) \leq R_{0-1}(h_{Gen,\infty}) + \varepsilon_0 \quad \text{with probability } 1 - \delta_0,$$

it suffices to take $m = O(\log n)$

Generalization error and minimal generalization error for loss ℓ :

$$R_\ell(h) = \mathbb{E}_{(x,y) \sim D} [\ell(h(x), y)], \quad R^* = \inf_{h \in H} R_\ell(h).$$

Definition ($\mathcal{H}$ -consistency bound)

Let ℓ_1, ℓ_2 be two loss functions. An $\mathcal{H}$ -consistency w.r.t. a hypothesis class H is a relation of the form

$$R_{\ell_2}(h) - R_{\ell_2, H}^* \leq f(R_{\ell_1}(h) - R_{\ell_1, H}^*) \quad \text{for all } h \in H,$$

for some non-decreasing function $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$.

If hypothesis has a small excess risk for surrogate-loss , then it will also have a small excess risk for true loss.

$\mathcal{H}$ -consistency

Notations

For ℓ define the pointwise infimum: $C_{\ell,H}^*(x) = \inf_{h \in H} C_{\ell,H}(x)$

$$C_{\ell,H}(x) = \sum_y P(Y = y \mid X = x) \ell(h(x), y),$$

The gap between $\mathbb{E}[C^*]$ and R^* : $M_{\ell,H} := R_{\ell,H}^* - \mathbb{E}_X[C_{\ell,H}^*(X)]$.

Pointwise and global optima misalign.

Theorem ($\mathcal{H}$ -consistency for binary logistic vs. 0–1)

Let $H_{lin} = \{x \mapsto \langle w, x \rangle + b : \|w\|_2 \leq W, |b| \leq B\}$ be the class of bounded linear predictors. If

$R_{\log}(h) - R_{\log,H_{lin}}^* + M_{\log,H_{lin}} \leq \frac{(e^B - 1)^2}{2(e^B + 1)^2}$, then for any distribution over (X, Y) the following $\mathcal{H}$ -consistency bound holds:

$$R_{0-1}(h) - R_{0-1,H_{lin}}^* \leq \sqrt{2} \cdot \sqrt{R_{\log}(h) - R_{\log,H_{lin}}^* + M_{\log,H_{lin}}}.$$

$\mathcal{H}$ -consistency Bound and Complexity for Multiclass Logistic Loss

Theorem (Multiclass $\mathcal{H}$ -consistency)

If $R_{\log}(h) - R_{\log, H_{lin}}^* + M_{\log, H_{lin}} \leq \frac{1}{2} \left(\frac{e^{2B}-1}{e^{2B}+K-1} \right)^2$, and for any distribution satisfying $\max_y p_y(x) - \min_y p_y(x) \leq \frac{e^{2B}-1}{e^{2B}+K-1}$, then

$$R_{0-1}(h) - R_{0-1, H_{lin}}^* + M_{0-1, H_{lin}} \leq \sqrt{2} \left(R_{\log}(h) - R_{\log, H_{lin}}^* + M_{\log, H_{lin}} \right)^{1/2}.$$

Theorem (Sample Complexity of Logistic Regression)

Suppose that ν satisfying $\nu < \frac{1}{2} \left(\frac{e^{2B}-1}{e^{2B}+K-1} \right)^2$. Let

$\varepsilon_0 \in [\sqrt{2\nu}, \frac{e^{2B}-1}{e^{2B}+K-1}]$ and $\delta_0 \in (0, 1)$. Then it suffices to take $m = O(n)$ training samples to guarantee, with probability at least $1 - \delta_0$,

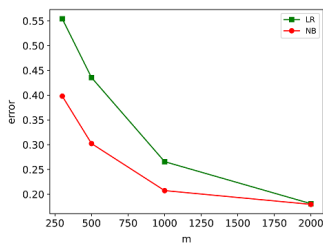
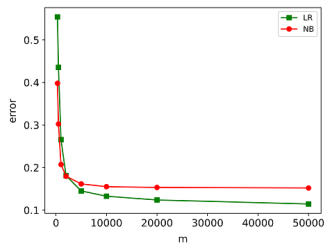
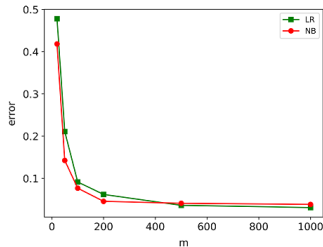
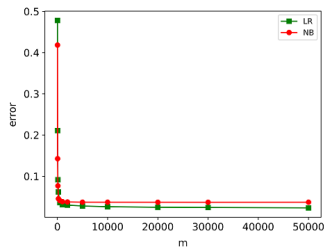
$$R_{0-1}(h_{Dis, m}) \leq R_{0-1}(h_{Dis, \infty}) + \varepsilon_0.$$

Convergence: Logistic Regression vs Naïve Bayes

Method	NB faster / CIFAR10	Two regimes CIFAR100
ViT	✓ / ✓	✓ / ✓
ResNet	✓ / ✓	✓ / ✓
CLIP	✓ / ✓	✓ / ✓
MoCov2	✓ / ×	✓ / ×
SimCLRv2	✓ / ×	✓ / ✓
MAE	✓ / ✓	✓ / ×
SimMIM	✓ / ×	✓ / ×

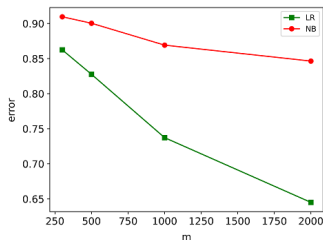
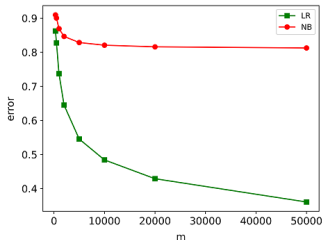
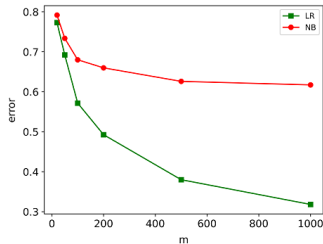
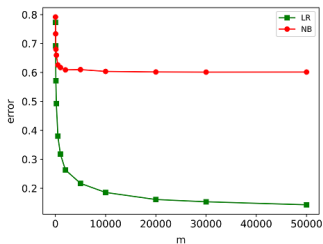
Convergence comparison between multiclass logistic regression and naïve Bayes. “NB faster” means naïve Bayes approaches its asymptotic error faster.

Convergence Curves



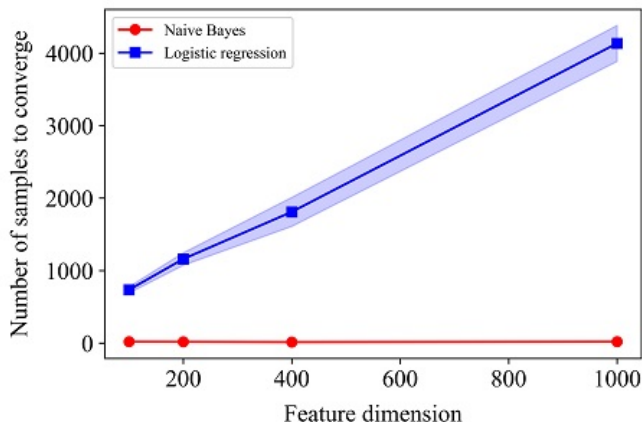
Comparison between naïve Bayes and logistic regression with the features extracted by ViT on the CIFAR10 (Above) and CIFAR100 (Below) dataset. M is a number of samples.

Convergence Curves



Comparison between naïve Bayes and logistic regression with the features extracted by ViT on the CIFAR10 (Above) and CIFAR100 (Below) dataset. M is a number of samples.

Convergence Curves



Multiclass ($K = 5$) simulation results. Empirically, logistic regression and naïve Bayes require $\mathcal{O}(n)$ and $\mathcal{O}(\log n)$ samples to approach the corresponding asymptotic error. Error bars show the variance estimated by 5 runs.

Conclusion

- ▶ Under mild assumptions multiclass Naïve Bayes needs only $\mathcal{O}(\log n)$ samples to approach its asymptotic error, while logistic regression requires $\mathcal{O}(n)$ samples.
- ▶ Experiments with diverse pre-trained vision models support theory and reveal the strength of the generative linear head in few-shot scenarios.